engineering 4 us all

Student Handout

Name: Stepan, Haralambos, Chloe, Sam **Date:** ________________ **Class:** _______________

MyDesign Portfolio

Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

The environmental science department needs a reliable and mold-free water system for their greenhouse to support healthy plant growth and maintain a sanitary learning environment. The current irrigation system retains moisture, leading to mold buildup that harms plant health and the efficiency of water distribution. A well-structured irrigation system will ensure even water distribution without stagnation, reducing the risk of mold and ensuring a healthier growing environment.

Prior Design Attempt #1 - PVC Pipes

Source

Gardener's Supply Company. "How to Choose a Watering System." 2023. https://www.gardeners.com/how-to/how-to-choose-a-watering-system/8747.html

Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

PVC pipes are widely used in irrigation due to their affordability, ease of installation, and corrosion resistance. They provide a simple way to distribute water evenly to plants.

Analysis

PVC is prone to biofilm and mold buildup, especially in humid environments like greenhouses. Additionally, PVC pipes can degrade over time under constant exposure to sunlight and fluctuating temperatures, leading to potential contamination of the water supply. This design does not fully meet the stakeholders' needs because it requires frequent maintenance to prevent mold buildup and clogging.

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Prior Design Attempt #2 - Drip Irrigation

Source

Water Technology Online. "Piping for Water and Wastewater Treatment Systems." 2023. https://www.watertechonline.com/wastewater/article/14210705/piping-for-water-and-wastewater-treatment-systems

Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Drip irrigation may also be a solution, as systems provide slow, consistent water delivery directly to the roots and are more efficient compared to the standard watering system in place currently, reducing water waste and minimizing mold-friendly standing water.

Analysis

These systems often suffer from clogging due to sediment and algae accumulation. Additionally, they require regular maintenance and do not fully prevent mold development in humid greenhouse conditions. This system would not fully meet stakeholders' needs because it does not address mold accumulation in the water lines.

Prior Design Attempt #3 - Hydroponic Systems

Source

Kellogg Garden. "How to Properly Water Your Garden with Water Systems." 2023. https://kellogggarden.com/blog/gardening/how-to-properly-water-your-garden-with-water-systems/

Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Hydroponic systems optimize water usage and minimize mold growth by using filtered and oxygenated water that circulates through the system.

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Analysis

These systems require costly filtration and oxygenation equipment to maintain water quality, making them impractical for a school greenhouse with limited resources. They also demand constant monitoring to prevent waterborne diseases. This design does not fully meet stakeholders' needs because of its complexity and maintenance requirements. They are also more suited for industrial uses, and therefore would be expensive to implement in the greenhouse.

Rubric

Element B: Documentation and Analysis of Prior Design Attempts						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B1. Documentation of plausible prior attempts to solve the problem and/or related problems is drawn from __________.	a wide array of clearly-identified and consistently-credible sources	a variety of clearly-identified and consistently-credible sources	several—but not necessarily varied—clearly-identified and generally-credible sources	a limited number of sources, some of which may not be clearly identified and/or credible	only one or two sources that may not be clearly identified and/or credible	missing or minimal (a single source that is not clearly identified and/or credible)
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B2. The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — __________.	is consistently clear, detailed, and supported by relevant data	is clear, generally detailed, and supported by relevant data	is generally clear and contains some detail and relevant supporting data	is overly general and contains little detail and/or relevant supporting data	is vague and is missing any relevant details and/or relevant supporting data	cannot be inferred from information intended as analysis of past and/or current attempts to solve the problem